

A Sense of Time for JavaScript and Node.js: First-Class Timeouts as a Cure for Event Handler Poisoning

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A Sense of Time for JavaScript and Node.js: First-Class Timeouts as a Cure for Event Handler Poisoning

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Abstract:

The software development community is adopting the Event-Driven Architecture (EDA) to provide scalable web services, most prominently through Node.js. Though the EDA scales well, it comes with an inherent risk: the Event Handler Poisoning (EHP) Denial of Service attack. When an EDA-based server multiplexes many clients onto few threads, a blocked thread (EHP) renders the server unresponsive. EHP attacks are a serious threat, with hundreds of vulnerabilities already reported in the wild.

We make three contributions against EHP attacks. First, we describe EHP attacks, and show that they are a common form of vulnerability in the largest EDA community, the Node.js ecosystem. Second, we design a defense against EHP attacks, First-Class Timeouts, which incorporates timeouts at the EDA framework level. Our Node.cure prototype defends Node.js applications against all known EHP attacks with overheads between 0% and 24% on real applications. Third, we

promote EHP awareness in the Node.js community. We analyzed Node.js for vulnerable APIs and documented or corrected them, and our guide on avoiding EHP attacks is available on nodejs.org.

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